ICH Harmonised Tripartite Guideline: Quality Risk Management (ICH Q9)

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1. Introduction and Scope

Risk management principles are effectively utilized across many areas of business, government, and industry. In the pharmaceutical sector, the protection of the patient is the utmost priority in manufacturing, development, and distribution.

This guideline provides principles and examples of tools for quality risk management that can be applied to different aspects of pharmaceutical quality. These aspects include development, manufacturing, distribution, and the inspection and submission/review processes throughout the lifecycle of drug substances, drug products, biological and biotechnological products.

A structured quality risk management approach enables clearer, more transparent decision-making, improves manufacturing efficiency, and facilitates regulatory compliance.

2. Principles of Quality Risk Management

Two primary principles of Quality Risk Management are central: The evaluation of the risk to quality should be based on scientific knowledge and ultimately link to the protection of the patient.

The level of effort, formality, and documentation of the quality risk management process should be commensurate with the level of risk.

Application of these principles fosters continual improvement and scientific understanding throughout the product lifecycle.

3. Quality Risk Management Process

Quality risk management is a systematic process for the assessment, control, communication, and review of risks to the quality of the drug product across the product lifecycle.

Key components include risk assessment (consisting of risk identification, risk analysis, and risk evaluation), risk control (including risk reduction and risk acceptance), risk communication, and periodic risk review.

Decision makers should ensure that stakeholders are adequately engaged in risk discussions.

4. Risk Assessment Methodologies and Facilitation

Common tools include Basic risk management facilitation methods (flowcharts, check sheets), Failure Mode and Effects Analysis (FMEA), Failure Mode, Effects and Criticality Analysis (FMECA), Fault Tree Analysis (FTA), Hazard Analysis and Critical Control Points (HACCP), and Hazard Operability Analysis (HAZOP).

The selection of the specific tool depends upon the complexity of the process and the availability of empirical data.